DNA Sequence Simulator Unit Test Table

Case	Test Group	Test Description	Total Assertions	Result
1	Base Simulator	Checks read structure, coverage maths, reverse complement, GC content, quality-to-error conversion, and profile loading/validation	18	✓
2	Homopolymer Detection	Checks that homopolymer runs are correctly detected, that all positions within a run are marked, and that the minimum length threshold is respected	7	✓
3	IO Readers	Checks that single, multi, and multiline FASTA files are parsed correctly; that headers and N bases are preserved; and that a missing file raises an error	9	✓
4	IO Writers	Checks that FASTA and FASTQ files are written with correct line counts, header format, Phred+33 quality encoding, and that a roundtrip preserves sequences and scores	15	✓
5	Illumina Batch	Checks that a batch produces reads with sequential IDs, that coordinates fall within sequence bounds, and that quality and sequence lengths match	5	✓
6	Illumina Quality Scores	Checks that scores match read length, never fall below the floor of 2, and that SSE and homopolymer penalties reduce quality on the correct strand only	5	✓
7	Illumina Error Injection	Checks output types, quality-correlated error rates, background/lagging/prephasing modes, sequence trimming on 3+ prephasing events, and weighted substitution distribution	11	✓
8	Illumina SSE Trigger Map	Checks that GGC/GCC motifs mark the correct forward/reverse positions within a 40 bp window, that boundary positions are respected, and that inverted repeats are detected	9	✓
9	PacBio Batch	Checks that batches produce reads with sequential IDs, valid coordinates, matching quality/sequence lengths, and that read lengths vary (log-normal sampling)	6	✓
10	PacBio Quality Scores	Checks that scores match read length, never fall below 2, that homopolymers reduce quality, and that an empty profile falls back to the Q38 baseline	4	✓
11	PacBio Error Injection	Checks all three error types (substitution, insertion, deletion), that output length reflects indel counts, that homopolymer errors are ∼95% deletions, and that background type ratios are approximately 22/45/33%	12	✓
12	Platform Contract	End-to-end integration test for both Illumina and PacBio pipelines; checks read structure, coordinate bounds, Phred range (2–40), coverage count, and that an error is raised when the sequence is shorter than the read length	16	✓